

Md. Mosharaf Hossain Apurbo

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🌐 [moshrf-hossain-apurbo](https://moshrf-hossain-apurbo.github.io)

📄 Roll-no-57

RESEARCH INTERESTS

Physical AI, machine learning, agentic AI workflows, and on-device / embedded machine learning.

EDUCATION

Bangladesh University of Engineering and Technology (BUET)

B.Sc. in Computer Science and Engineering (Level-4)

Advisor: [Dr. Atif Hasan Rahman](#) (Associate Professor, Dept. of CSE, BUET)

- Relevant courses: Signals and Systems, Computer Architecture, Data Structures and Algorithms, Theory of Computation, Software Engineering, Database, Object-Oriented Programming.

Notre Dame College

Higher Secondary Certificate (HSC), GPA 5.00/5.00

Dhaka, Bangladesh

Nov 2022 – Present

Dhaka, Bangladesh

Jul 2019 – Dec 2021

RESEARCH EXPERIENCE

Generating Scale-Free Chordal Graphs

Undergraduate Thesis, BUET — Supervisor: [Dr. Md. Saidur Rahman](#) (Professor, Dept. of CSE)

Ongoing

Jun 2026 – Present

- Exactly characterised which degree distributions are achievable by chordal graphs of bounded clique number, and showed the same condition holds for k -degenerate, bounded-treewidth and bounded-pathwidth graphs.
- Designed a **linear-time generator** that outputs chordal graphs with a prescribed degree distribution (with a perfect elimination order as certificate); for power laws it attains the **optimal convergence rate** with no degree cut-off.

SpatialNet: Power-efficient, Effortless Onboarding with Smart Glass

CSE-450 Capstone, BUET — with [Samsung R&D Institute Bangladesh](#)

Under review

Dec 2025 – Sep 2026

Md. Mosharaf Hossain Apurbo, Niloy Das Robin, Nabiha Tahseen, Gourave Roy, Sayjad Rahman

- A **0.72M-parameter (1.5 MB)** model that finds the wearer's hand, fingertip and pointed-at target from a single frame, running fully on a **Samsung WearOS-based smart glass** (60–80 ms/frame).
- **82.6% target IoU** and **99.2% scene accuracy**; +34 pts IoU over vision-language models with 5,000× fewer parameters.

PUBLICATIONS

1. Md. Mosharaf Hossain Apurbo, Md. Asif Ali, and Kazi Reyazul Hasan, **Multiple-Instance Contrastive Alignment for Pose-Based Sign Spotting**, Workshop on Sign Language Processing (WSLP) at EMNLP, 2026. [Workshop Accepted](#)
2. Md. Asif Ali, Md. Mosharaf Hossain Apurbo, and Kazi Reyazul Hasan, **Metric-Learning Retrieval for Cross-Signer Isolated Sign Language Recognition**, Workshop on Sign Language Processing (WSLP) at EMNLP, 2026. [Workshop Accepted](#)

SELECTED PROJECTS

Trip.AI — AI Travel Planning Platform 🤖

Oct 2024

Agentic AI application

- Built **agentic, personalised itinerary generation** that plans trips around each user's preferences, with real-time weather updates.
- Implemented **RAG-based natural-language image search** during the BUET CSE Fest Hackathon 2024 (Best Documentation award).
- Added AI-generated travel blogs and vlogs with accompanying music.

Mini-PC: 8-bit MIPS Processor (Logisim) 🤖

Oct 2024

Computer architecture

- Designed an 8-bit MIPS processor from scratch: control logic, ALU, memory access and I/O for a complete instruction cycle.
- Built a display module and created a **Snake game** that runs on the processor.

C-to-Intel-8086 Compiler (C++17, ANTLR 4)

Jan 2025 – Jun 2025

Low-level systems programming

- Full front end (lexer, semantic analysis, scoped symbol table) and 8086 back end with BP-relative stack frames, recursion and arrays; peephole optimiser.

Quantum Machine Learning Benchmarking for Biomedical Data

Scientific ML

- Benchmarking framework for quantum feature encodings and ansatz models with SHAP/LASSO/PCA feature selection and convergence visualisations.

BookSphere — E-commerce Platform

Jan 2024 – Feb 2024

Full-stack web application

- An Oracle-powered, Bootstrap-based e-commerce site replicating Rokomari.com's core features.

HONOURS

- 2nd Runners-up, AI Proof of Concept Exhibition, Samsung R&D Institute Bangladesh, Sep 2026.
- Best Documentation (AI/API), BUET CSE Fest Hackathon 2024.

SKILLS

Programming	C, C++, Python, Java, SQL
ML	PyTorch; on-device model design, quantisation and compression for resource-constrained hardware; contrastive and metric learning
Systems	Embedded C, computer architecture, compilers; Git, Docker
Other	Next.js, Spring Boot; Oracle, MySQL, MongoDB